

Pune Intersection Simulation — Structured Analysis (Who / What / When / Where / Why / How)

A deeper breakdown of the case-study document, organized so it can be used as a standalone reference alongside the Monday prep sheet.

1. WHAT — What is being built, and what problem does it solve

What it is: A configurable traffic-intersection simulator whose default behavior is calibrated against Indian (specifically Pune) urban intersections rather than the symmetric, homogeneous-fleet assumptions baked into Western traffic engineering standards (HCM).

What breaks the standard model:

Symmetric HCM models assume equal lanes in/out, homogeneous vehicle types, and lane discipline. Pune intersections violate all three simultaneously.

Core mismatch: Western micro-simulators treat "lanes" as hard channels vehicles queue inside. Pune traffic treats lane markings as advisory — two-wheelers fill gaps between and around queued cars ("porous flow" / "lane seepage").

What the deliverable actually contains:

Component	Purpose
4 real-world case studies (SPPU Chowk, Nal Stop, Sancheti Chowk, Swargate)	Ground the model in observed, physically distinct failure modes
A pattern → mechanism mapping table	Translates each observed behavior into a specific simulation rule
A parameter/bounds matrix	Gives the engine tunable knobs with min/max/default values
A Monday meeting checklist	Converts open design questions into decisions to be ratified with Khushi

2. WHY — Why these specific intersections, why this approach

Why 4 intersections and not 1: Each was chosen to isolate a *different* structural failure mode, not just to add data points:

- SPPU Chowk** → tests **multi-leg asymmetry + directional tidal reversal** (5 legs, 80:20 AM/PM flip).
- Nal Stop** → tests **vertical/physical lane constriction** (3-lane-to-2-lane drop under metro piers) plus **informal capacity loss** (buses/autos parking in the running lane).
- Sancheti Chowk / JM-FC Road** → tests **strict one-way corridor pairs** feeding into a shared conflict zone with no opposing traffic on the arterials themselves but heavy cross-conflict at the junction.
- Swargate** → tests **fleet heterogeneity friction** — wildly different acceleration/turning envelopes (12m bus vs 100cc bike) sharing a stop line and turn radius.

Why this matters for the simulator specifically: If the engine only handled one of these patterns, it would model *a* Pune intersection, not *Pune-style* intersections generally. The four cases were selected so their union covers the parameter space the config matrix later formalizes (lane asymmetry, one-way logic, fleet mix, turn friction).

Why deviate from HCM defaults: HCM's queuing and gap-acceptance formulas assume lane-disciplined, homogeneous vehicles arriving in platoons. Under bike-seepage and porous flow, the effective saturation flow rate and critical gap behavior are structurally different — hence the need for a distinct "Pune Preset" fleet mix and a stochastic aggressiveness factor rather than fixed HCM constants.

3. WHO — Who is affected / involved in each behavioral pattern

Actor	Role in the chaos	Where it shows up
Two-wheeler riders	Filter through stopped traffic to form a front swarm at red lights	SPPU Chowk "bike box"; general porous filtering pattern
Auto-rickshaw & PMPML bus drivers	Pull over inside the intersection approach, cutting usable lanes	Nal Stop (3 lanes → 1.5 effective lanes)
Heavy vehicle drivers (buses/trucks)	Wide turning radius blocks all other movements when turning right	Swargate (Shivaji Rd → Shankarsheth Rd)
Commuters on IT-corridor routes	Drive the 80:20 directional tidal split	SPPU Chowk AM/PM reversal (Baner/Aundh ↔ Shivajinagar)

Actor	Role in the chaos	Where it shows up
Pedestrians	Cut across traffic streams, adding an uncontrolled conflict class	Swargate
Khushi (meeting counterpart)	Co-owner of the design decisions on lane rules, phasing, fleet defaults, turn splits	Section 4 checklist
Engineering team (implied "we")	Owns <code>network.py</code> , <code>router.py</code> , <code>spawner.py</code> implementation once parameters are locked	Suggested Next Steps

4. WHEN — Temporal dynamics that must be modeled

Peak-hour reversal: SPPU Chowk flips its dominant flow direction between AM peak (inbound to city) and PM peak (outbound to Hinjawadi/Baner) — this is not a static `arrivalRate`, it needs a **time-of-day-dependent** arrival rate per approach.

Signal-cycle timescale: Cycle length (30–180s), green/yellow/all-red phases operate on a much faster clock than the tidal shift, and must be independently configurable per direction (major vs minor street).

Sub-second driver behavior: "Sneaking" — vehicles clearing the intersection 1–2 seconds into yellow/red — happens at the granularity of the simulation's physics tick, driven by the stochastic `driverAggressiveness` factor.

Ongoing infrastructure state: Nal Stop's metro construction is a semi-permanent geometric constraint (not a real-time variable), but worth flagging as a *scenario preset* (e.g., "under-construction" lane profile) rather than a live parameter.

5. WHERE — Spatial/structural loci of failure

At the stop line: Bike-box swarming (SPPU), fleet launch-speed mismatch (Swargate).

Mid-approach, before the box: Weaving conflicts within short distances (40m weave at SPPU Chowk), informal parking-lane loss (Nal Stop).

At the physical pinch point: Metro pillars/flyover piers reducing lane count independent of signal design (Nal Stop).

At the confluence, not the corridor: One-way pairs (JM/FC Roads) have zero conflict along the arterial itself; all conflict concentrates at Sancheti Chowk where multiple one-ways merge.

Across the whole approach geometry: T-junctions and closed arms are a *topology* concern (Section 3 "Approach Topology Modes"), separate from lane-count or signal-timing concerns.

6. HOW — How each real-world pattern becomes a simulation mechanism

This is the direct translation table from the source document, restated as cause → engine rule:

Porous filtering → shorter vehicle classes are permitted to bypass queued larger vehicles up to the stop line (a queue-model exception, not a new lane).

Tidal directional split → `arrivalRate` becomes a per-approach, independently settable parameter (not a single global rate).

One-way corridors → `incomingLanes` and `outgoingLanes` can independently be set to 0 per approach, with a downstream rule (open question in Section 4) on whether `incoming = 0` auto-disables the signal head/spawner for that arm.

Fleet heterogeneity → vehicle-type distribution becomes configurable, each type carrying distinct length, width, and IDM (Intelligent Driver Model) acceleration bounds.

Asymmetric phase allocation → green duration is per-direction rather than a single global `greenDuration`.

Free left turns → an optional slip-lane flag lets left-turning vehicles bypass the signal conflict manager entirely.

Aggression/sneaking → a stochastic `driverAggressiveness` factor perturbs both the critical gap (t_c) and how far into yellow/red a driver will still commit to crossing.

How the parameters are bounded (engineering safeguard): every tunable (lanes, cycle length, gap times, fleet %) has an explicit min/max/default, so the UI can't be pushed into physically nonsensical states (e.g., a lane width below 2.5m or a cycle shorter than 30s).

How topology choice constrains lane values: the four "Approach Topology Modes" (Two-Way, One-Way Inbound, One-Way Outbound, Closed/T-Junction) are really just named presets over the same two underlying sliders (`incoming`, `outgoing`), which keeps the parameter model small while covering all the real intersections described.

7. WHICH — Which decisions are still genuinely open (not yet resolved by the analysis)

These are the items where the document explicitly defers to Monday's meeting rather than asserting an answer — worth flagging as the real decision surface:

Does `incoming = 0` **automatically** disable the signal head and vehicle spawner for that arm, or can a phantom/inactive arm still hold a signal head for visual completeness?

Is a 5-lane-per-direction ceiling actually sufficient, or should highway-scale approaches (which the doc gestures at) get a higher ceiling?

Two-phase (N-S / E-W) vs four-phase signal control, and whether a **protected right-turn phase** is a global option or only for flagged high-volume junctions like Nal Stop/Law College Rd.

Whether vehicle-type physical properties (length, acceleration, top speed) are **fixed presets** per class or exposed as **advanced sliders** — this is a UX/complexity trade-off, not just a modeling one.

Default turn-ratio splits differ by junction type (70/15/15 for arterials vs 50/50 for T-junctions) — confirm whether this should be auto-selected based on topology or manually set.

8. HOW MUCH — Quantified stakes / calibration targets

Directional imbalance as extreme as **80:20** at SPPU Chowk during peak hours — this is the outer bound the `arrivalRate` asymmetry needs to support, not a typical case.

Effective capacity loss at Nal Stop: **3 lanes reduced to ~1.5 usable lanes** due to informal stopping — meaning the simulator's "lane count" parameter alone is insufficient; an informal-capacity-loss modifier may be needed to match reality even when nominal lanes are set to 3.

Pune fleet-mix baseline proposed: **50% two-wheelers / 15% autos / 30% cars / 5% heavy vehicles**, contrasted against a Western/HCM preset of **5/0/85/10** — this contrast is the single clearest justification for why a "Pune Preset" needs to exist as a first-class default rather than a tweak of a Western default.

Output metrics proposed for validating the model against reality: average delay (sec/vehicle), LOS grade (A-F), signal-vs-roundabout crossover volume (veh/hr), and max queue length — these are the yardsticks by which "does this simulation actually reflect Pune" will be judged.

9. Gaps Worth Raising Beyond the Current Document

No mention of **pedestrian volume** as a configurable parameter, despite Swargate explicitly calling out pedestrian streams as a conflict source — currently this seems to be flavor text rather than a modeled input.

No parameter for **informal lane loss** (parked autos/buses reducing effective lanes) — Nal Stop's 3→1.5 lane example is described narratively but doesn't yet map to an entry in the Section 3 config matrix.

The aggressiveness factor is described as adjusting t_c and yellow-clearance behavior, but its **distribution shape** (uniform? normal? per-vehicle-type?) isn't specified — worth deciding before implementation, since it affects both realism and reproducibility of results.